

Granular Impact Mechanics with Jerboa-Inspired Feet: Reporting, Figures, Area, Scaling, and Dynamic-Force Analysis Against the Katsuragi-Durian Framework

v2 — reviewer edit applied: KD Fig. 1 insets deferred until after the dynamic study.

TL;DR

Test KD as-is and report a single pooled measured- v_0 range with per-height mean ± 1 SD as the x-coordinate. The KD lineage (Katsuragi & Durian 2007; Ambroso et al. 2005; Newhall & Durian 2003) plots against measured impact speed (or total drop distance $H = h + d$), averages replicates per condition, and routinely pools projectiles/shapes onto shared axes — so a single v_0 range across your three same-mass geometries is defensible and standard practice.

For area, follow your PI in using the constant convex-hull area for the depth (hydrostatic, k) term, but treat the inertial (Cv^2) term with a physically motivated effective area and report both bounds. At your toe-gap-to-grain ratio (~ 59 – 118 grain diameters) grains flow freely through the gaps — 10 – $20\times$ above every known granular bridging/clogging threshold (~ 3 – 6 grain diameters) — so summed bare-element area is the physically correct inertial reference and the convex hull is an upper bound.

Test the depth-velocity $2/3$ law on linear axes (the PI is right), pairing a fixed- $2/3$ fit with your own free-exponent fit (CI vs the $2/3$ target) and an area-normalized comparison across geometries — not a log-log power-law fit over your <1 -decade range; the compensated plot is optional. Present the dynamic KD analysis as a three-step sequence (fixed-depth a -vs- v^2 sweeps $\rightarrow F(z)$ collapse $\rightarrow$ speed-independence + a_{stop} discontinuity), replicated per geometry.

Key Findings

Item 1 — Audit and velocity binning

KD 2007's driving variable is **measured** instantaneous speed. As the paper states verbatim, "we developed a means to measure projectile dynamics with 100 nm and 20 μs precision. For a 1-inch-diameter steel sphere dropped from a wide range of heights into non-cohesive glass beads..." v_0 is extracted from a fit to the imaged velocity trace, and the paper explicitly notes its imaged v_0 and d are "consistent with, but more accurate than" the $\sqrt{2gh}$ and micrometer-gauge values. This directly validates your imaging-based v_0 (rail friction makes $\sqrt{2gh}$ an overestimate).

KD 2007 reports a **single continuous span of impact speeds "from 0 to ~ 400 cm/s"** across all drops on one figure (Fig. 1), color-coded by v_0 , rather than per-condition sub-ranges. It uses one projectile so there is no cross-shape pooling, but the convention is a single pooled speed axis.

Ambroso et al. 2005 and Uehara et al. 2003 plot penetration depth d against **total drop distance $H = h + d$** and against v_0 , pooling many spheres of different diameter/density onto shared axes; the collapse across projectiles is the central result. Newhall & Durian 2003 similarly pool cylinders of varied length, diameter, density and tip shape onto common scaling axes.

Replicate/error-bar practice in the lineage: the closest direct methodological analog, Williams 2025 ("Geometry-controlled Onset of Inertial Drag in Granular Impact," arXiv:2512.22781, OIST), states verbatim: "For each height, at least five drops were performed... the mean across all drops was reported as the measured value, with vertical error bars corresponding to the standard deviation," with horizontal error bars from drop-height uncertainty. This **standard-deviation (not standard-error)** convention for spread across replicate drops is the field norm.

Recommendation: State a single pooled measured- v_0 range in text (e.g., "measured v_0 spanned X–Y m/s across all three geometries and 19 drop heights"). Use the **per-height mean measured v_0 as the representative x-coordinate**, with **± 1 SD horizontal error bars** representing drop-to-drop spread (rail friction). Because all three feet share mass, material, drop heights and conditions, per-geometry ranges will overlap almost completely; separate ranges would falsely imply a systematic difference you do not have. Report per-geometry ranges only in a supplementary table if reviewers ask. Use SD for spread of replicates; if you additionally plot a mean-of-means summary, SE is acceptable but must be labeled explicitly.

Item 2 — Kinematics figures (KD Fig. 1 layout)

KD 2007 Fig. 1 is a 1×3 vertical stack (three stacked panels, shared time axis), not side-by-side: (a) depth $z(t)$, (b) speed $v(t)$, (c) net acceleration $a(t)+g$ vs time t , all curves color-coded by initial impact speed v_0 , with many drop heights overlaid on each panel. Note KD plots **$a+g$** (net acceleration, subtracting gravity) rather than raw a . For reference, KD's insets are Fig. 1b inset = stopping time t_{stop} vs v_0 (with characteristic time/velocity scales) and Fig. 1c inset = $a+g$ vs depth for the $v_0=0$ case compared to the kz Coulomb-friction expectation — but **defer replicating these insets for now**: they draw on quantities (t_{stop} scales, the kz comparison) that belong naturally to the dynamic study, so add them after Step 5 if at all.

Origin/sign convention: $t=0$ at initial impact, and **keep your own pipeline convention — depth measured positive into the bed** (as `kd_kinematics` outputs it), rather than adopting KD's upward-positive z . For the record, KD 2007 plot z upward-positive (penetration negative), so your depth curves will appear mirrored relative to theirs; a one-line caption note ("depth positive into the bed") handles the mapping. The qualitative feature the panels are built to expose is unchanged either way: the **abrupt velocity vanishing with an acceleration discontinuity at stoppage** — counter to a viscous (fluid) approach.

Comparable multi-condition papers: Goldman & Umbanhowar 2008 present acceleration-centric kinematics emphasizing the "two jumps" (velocity-dependent rise at contact; depth-dependent drop at rest); Katsuragi & Durian 2013 plot **v vs z** (phase-space) as the primary kinematic view because the KD ODE has a closed-form solution in v - z space (Ambroso, Kamien & Durian 2005). Newhall & Durian's cross-shape comparison is done on collapsed scaling axes rather than raw kinematics.

Marker scheme: Open/filled/crossed symbols to distinguish conditions are standard and APS-acceptable, but your proposed scheme (Default = filled circle ●, Tight = hollow circle with × ⊗, Wide = hollow circle ○) risks the ⊗ being confused with ● at small sizes and degrading in grayscale print. APS/PRE figure guidelines require legibility in grayscale and encourage colorblind-safe design, with symbol shape and fill (not color alone) carrying the distinction. **Recommendation:** Use three clearly distinct shapes **and** fills — e.g., Default = filled circle ●, Tight = open square □, Wide = open triangle △ — plus a colorblind-safe hue per geometry (Okabe-Ito/Wong palette: black, orange, sky-blue). This is more robust than three circles differing only by fill/overlay.

Recommended panel content for your 1×3: One panel per geometry (Default, Tight, Wide). Two defensible options: (A) mirror KD exactly — within each geometry panel overlay $z(t)$, $v(t)$, $a(t)+g$ for all drop heights (data-dense, directly comparable to KD Fig. 1), best realized as a 3×3 grid (rows = z , v , $a+g$; columns = geometries); or (B) if space-limited, plot **v vs z per geometry** (KD 2013 style) since it best exposes the fitted k and C . Given the "test KD as-is" directive, option A most faithfully replicates KD; the compact 1×3 v -vs- z panels (drop heights color-coded) are the strong fallback. Keep the $a+g$ (net acceleration) convention to match the literature, use your own depth-positive-into-the-bed sign, and declare both in captions.

Item 3 — Area verification (convex hull vs bare vs effective)

The KD inertial term is written, verbatim in Williams 2025 Eq. 1, " $F(z,v) = F_{qs}(z) + C_{p_g} A v^2$," with " A the effective area of the projectile and ρ_g the grain density." The depth term $F(z) \approx kz$ is hydrostatic/Coulomb-like, and in KD 2007 k is expressed via the ball area/diameter (characteristic length $L_c = D_b$, k on the order of $\rho_g g D_b^2$ per unit). So **area enters both terms, and the two terms need not share the same effective area** — a key point for splayed feet.

Gap-vs-grain physics is the crux. Your toe gaps (~ 5.9 – 11.8 mm) are ~ 59 – $118\times$ the ~ 0.1 mm grain diameter. Across the granular literature, bridging/jamming between elements requires gaps of only a few grain diameters:

Parallel-rod intrusion (Pravin et al., Phys. Rev. E 104, 024902, 2021): verbatim, "We observed a peak in W at a gap spacing of ~ 3 particle diameters, which was up to 25% greater than W at large separation (>11 particle diameters), beyond which the total work plateaued." $\rightarrow$ cooperative jamming peaks at ~ 3 grain diameters and vanishes (elements act independently) beyond ~ 11 .

3D silo clogging ceases above outlet apertures of ~ 5 grain diameters (Zuriguel et al. 2005, Phys. Rev. E 71, 051303; Janda et al. 2008, EPL 84, 44002).

Proppant bridging across perforations requires aperture $< \sim 6$ grain diameters (Gruesbeck & Collins 1982).

At 59–118 grain diameters you are 10–20× above every bridging threshold, so grains flow freely through your toe gaps — there is no stable stagnant bridge spanning them.

Implication for the inertial term: **the bare (summed material) frontal area is the physically correct reference for Cv^2** (momentum transfer is to grains actually struck by material). Resistive Force Theory (Li, Zhang & Goldman 2013, Science 339:1408) formalizes this — net force is the **linear superposition of per-element forces**, each scaling with that element's own plate area, valid in the quasi-static/frictional-fluid regime (stated validity $\leq \sim 0.5$ m/s for 0.3-mm glass beads) — consistent with summed bare area when elements are widely spaced.

Implication for the depth term: the granular Archimedes framework (Kang, Feng, Liu & Blumenfeld, Nat. Commun. 9, 1101, 2018) defines resistance via the intruder **cross-section S** and applies to convex shapes; verbatim, "the gradient of the steady-state part, $K\phi$, depends only on the medium's internal friction angle, ϕ , and... is nonlinear in $\mu = \tan \phi$... the intrusion of any convex solid shape satisfies a modified Archimedes' law." As the crater/stagnant zone develops around a splayed foot at depth, the effective quasi-static footprint can approach the convex hull. So the PI's convex-hull choice is best supported for the **depth term**, while bare/effective area is best for the **inertial term**.

Recommendation: (1) Use the **constant convex-hull area** as the single normalizing area for cross-geometry comparison of the **k (depth) term** — geometry-defined, constant, and consistent with the quasi-static/Archimedes convex-shape framework. (2) For the **inertial term**, report the fitted lumped parameter d ,

(where $mv^2/d_i = C \rho_g A v^2/m$) **without forcing an area assumption**, then separately quote the implied C under both bare and convex-hull A as bounding cases. (3) Explicitly state the gap/grain ratio and the free-flow conclusion, citing Pravin et al. 2021 and the clogging thresholds. (4) Flag the area ambiguity as a named limitation and an opportunity: the depth and inertial terms plausibly reference different effective areas, and your three splay geometries are precisely the instrument to test this.

Item 4 — Scaling analysis (2/3 law on linear axes, own exponent, area-resolved geometry comparison)

Ambroso et al. 2005 / Uehara et al. 2003 establish $d \propto \rho_b^{1/2} D_b^{2/3} H^{1/3}$, equivalently $d \propto v_0^{2/3}$ in the velocity domain ($H \propto v_0^2$ for the free-fall component). They present this via collapse across projectiles, with exponents reported as clean fractions (1/3 in H , 2/3 in v_0).

The PI is statistically correct to prefer linear axes over a log-log power-law fit. Your depth range (~1–3 cm) spans well under one decade. The power-law-fitting literature (Clauset, Shalizi & Newman 2009, SIAM Rev. 51:661; Stumpf & Porter 2012) warns that least-squares slopes on log-log axes over <1–2 decades of dynamic range are biased, under-report uncertainty, and can yield high R^2 even when the data are not a power law. Over <1 decade a fitted exponent cannot sharply discriminate 2/3 from nearby values — which shapes *how* to present your own exponent, not whether to fit it.

Own fitted exponent vs 2/3 (per the PI): fit the free exponent n per geometry alongside the fixed-2/3 fit, and present them together — 2/3-fixed as the theory-driven primary fit ($d = d_0 + \beta v_0^{2/3}$ or the $(d_0^2 H)^{1/3}$ form), the free n with its full confidence interval as the direct test of whether 2/3 is *consistent* with each geometry's data. This is exactly the structure your `depth_scaling` pipeline already emits ("Free exponent, cross-check (target 2/3)" with CIs; the sample run gave $n = 0.63 [0.50, 0.76]$ for Default GB/full, 2/3 in CI). Report it as "2/3 lies within the CI for all/some geometries," not as a measured exponent claim.

Area-resolved geometry comparison (per the PI): bring the Item 3 areas into the scaling itself. The Ambroso/Uehara law carries $D_b^{2/3}$, i.e., a characteristic-size dependence — for non-circular intruders the natural generalization is an area-based length $L_A = \sqrt{A}$. Plot d normalized by $L_A^{2/3}$ (equivalently $A^{1/3}$) vs v_0 for the three geometries, once with $A =$ convex hull and once with $A =$ bare material area: if the three curves **merge** under one area choice and **split** under the other, the scaling figure itself identifies which area governs penetration depth — turning the geometry comparison into a result rather than a caveat, and directly linking Steps 3 and 4.

Presentation options (linear axes throughout): (a) **d vs v_0** with fitted 2/3 curves per geometry — KD 2007 Fig. 2's own approach; (b) **area-normalized $d/A^{1/3}$ vs v_0** testing which area collapses the geometries; (c) **collapse d/d_0 vs v_0/v^*** onto one master curve; (d) *optional*: the compensated plot $d/v_0^{2/3}$ vs v_0 (flat iff 2/3 holds) — informative but not necessary given (a)–(c) and the free-exponent CI cover the same question; keep it in reserve for supplementary material or reviewer response.

Recommendation — 3 figures for the scaling section: (1) **d vs v_0 (linear axes)**, three geometries with fixed-2/3 fitted curves, per-height mean $v_0 \pm 1$ SD, and the per-geometry free exponent n [CI] quoted in the caption or an inset table against the 2/3 target. (2) **Area-normalized $d/A^{1/3}$ vs v_0** under both area definitions (convex hull vs bare), showing which one collapses the three geometries. (3) **Collapse d/d_0 vs v_0/v^*** across geometries. Fit d_0 and the prefactor with the exponent fixed at 2/3 for the primary curves; report goodness-of-fit as residuals on linear axes, not R^2 on log-log.

Item 5 — Dynamic study (KD force-law fitting)

KD 2007's dynamic analysis is a three-step decomposition (their Fig. 3 plus Fig. 1 insets):

Fixed-depth velocity sweeps (Fig. 3a): at five fixed depths $z_i = \{0, -1, -2, -3, -4\}$ cm, plot $a+g$ vs v (and vs v^2 in the inset). The data are quadratic in v with the **same proportionality d_1 (≈ 8.7 cm) at all depths** — the inset lines become parallel vs v^2 , proving the inertial term mv^2/d_1 is depth-independent. This isolates C/d_1 .

F(z) collapse (Fig. 3b): compute $F(z)/m = a + g - v^2/d_1$ and plot vs depth; all drop heights collapse onto one speed-independent friction curve. KD show it is well fit by the linear form $F(z) = kz$ ($k/m = 1040 \pm 10 \text{ s}^{-2}$), better than the saturating form over most of the range. This isolates k .

Consistency checks: stopping time t_{stop} vs v_0 (Fig. 1b inset) plateaus at high speed at the characteristic scale $(D_b/g)^{1/2}$; and the **acceleration discontinuity at stop** $\Delta a = g + (k/m)d$ (Fig. 1c) is predicted directly by the force law.

Follow-ups: Goldman & Umbanhowar 2008 emphasize the **two acceleration jumps** (velocity-dependent jump at contact; depth-dependent drop at rest) and warn additive separability fails away from the critical packing state; Umbanhowar & Goldman 2010 (Phys. Rev. E 82, 010301) show **force is cleanly separable into (depth-linear + velocity-quadratic) only near ϕ_{cps}** — the quadratic drag coefficient's depth-dependence changes sign across ϕ_{cps} . At $\phi = 0.624$ (fairly dense) you should verify separability rather than assume it.

Takizawa/Katsuragi and Okubo & Katsuragi 2022 fit the same ODE $m \ddot{z} = mg - mv^2/d_1 - kz$ and report (d_1, k) tables per condition. KD 2013 fits a **single (k, d_1) pair simultaneously across all impact speeds** in v -vs- z space, warning that per-speed-adjusted k, d_1 "do not represent well-defined material parameters." That single-pair-per-material protocol is the gold standard.

Standard validation tests: the speed-independence test (plot $a + g - v^2/d_1$ vs z and confirm collapse across v_0); depth-independence of d_1 (parallel lines in $a+g$ vs v^2); and a_{stop} vs depth (discontinuity magnitude should follow $g + (k/m)d$).

Recommendation — analysis sequence and 3 figures:

Fig. D1 (per geometry, KD Fig. 3a analog): $a+g$ vs v at fixed depths, inset vs v^2 , demonstrating quadratic-in- v and constant d_1 . Extract d_1 ($\rightarrow C$) per geometry.

Fig. D2 (per geometry, KD Fig. 3b analog): $F(z)/m = a+g-v^2/d_1$ vs z collapse across all drop heights, fit k . Overlay the three geometries' $F(z)$ curves to compare k directly.

Fig. D3 (cross-geometry summary): fitted k and C (or d_1) as scatter/bar with uncertainty for Default/Tight/Wide, plus an a_{stop} -vs-depth panel showing the discontinuity $\Delta a = g + (k/m)d$ per geometry. Fit one (k, d_1) pair per geometry simultaneously across all speeds (KD 2013 protocol), not per drop. Because you image kinematics (not an embedded accelerometer), differentiate $v(t) \rightarrow a(t)$ with the KD windowed line-segment fitting to keep acceleration uncertainty below $\sim 0.5\%$ or $0.005g$, and note that imaging cannot resolve the fast force-chain fluctuations that Goldman & Umbanhowar and Clark et al. capture.

Details

Fitting protocol nuance. KD's closed-form solution in v - z space (Ambroso, Kamien & Durian 2005; used by KD 2013) lets you fit $v^2(z)$ analytically for the ODE $m \ddot{z} = mg - kz - mv^2/d_1$, fitting (k, d_1) to the entire v -vs- z trajectory of each drop and then requiring a **single shared (k, d_1)** across all drop heights for a given geometry. This is more robust than numerically differentiating to $a(t)$ and is the recommended primary estimator; the fixed-depth a -vs- v^2 view (Fig. D1) then serves as a visualization/validation of depth-independence rather than the primary fit.

Packing-fraction caveat specific to your ϕ . Umbanhowar & Goldman 2010 found additive term separation is exact only near ϕ_{cps} ; away from it the quadratic drag coefficient acquires depth dependence (decreasing with depth below ϕ_{cps} , increasing above). Since the directive is to test KD as-is, explicitly report a separability check (does d_1 stay constant across depths in Fig. D1?) as evidence for or against clean KD applicability at $\phi = 0.624$.

Cross-geometry logic. Your same-mass design means differences in fitted k and C across Default/Tight/Wide are attributable to geometry (splay), not mass — a clean contrast. If the convex-hull envelope is designed similar across the three splays, it is precisely the right common normalizer: it isolates the effect of how material is distributed within a fixed envelope, testing whether k scales with the envelope (convex hull) or with the material footprint (bare area). A publishable outcome is a **split**: k tracking convex-hull area while C tracks bare area (or vice versa).

Geometry-dependent exponent as an alternative story. If the compensated plot is not flat, the $2/3$ law does not hold uniformly across splay, and a geometry-dependent crossover becomes the finding — directly paralleling Williams 2025, whose crossover speed to inertial drag "scales approximately linearly with the cone angle through $\tan \phi$," and whose peak force "collapses when rescaled by $\tan \phi$." For splayed feet, an analogous splay-angle rescaling would be the natural test.

Recommendations (staged)

Now (reporting/figures): Adopt a single pooled measured- v_0 range in text; per-height mean $v_0 \pm 1$ SD as x-coordinate. Build kinematics figures on the KD 1×3 template with a+g net-acceleration convention, your own depth-positive-into-the-bed sign, drop heights color-coded, and a colorblind-safe distinct-shape marker scheme ($\bullet$ / $\square$ / $\triangle$). Defer the KD-style insets (t_{stop} vs v_0 ; a+g vs depth) until after the dynamic study.

Scaling section: Present (i) d vs v_0 on linear axes with fixed- $2/3$ curves plus per-geometry free exponent n [CI] vs the $2/3$ target, (ii) area-normalized $d/A^{1/3}$ vs v_0 under convex-hull and bare areas to show which collapses the geometries, (iii) collapse d/d_0 vs v_0/v^* . Do not fit a power law on log-log axes over <1 decade; keep the compensated plot as optional/supplementary.

Dynamic section: Fit one (k, d_1) per geometry across all speeds via the v -vs- z closed form; validate with fixed-depth a -vs- v^2 (constant d_1) and $F(z)$ collapse; present k, C , and a_{stop} across geometries. Revisit the deferred kinematics insets here if desired.

Area: Normalize the k term by convex-hull area; report C under both bare and convex-hull area as bounds; state the gap/grain free-flow argument (Pravin et al. 2021; Zuriguel et al. 2005; Gruesbeck & Collins 1982).

Benchmarks that change the recommendation: If the separability check (Fig. D1) shows d_1 drifting systematically with depth, KD-as-is is insufficient at your ϕ ; report the ϕ_{cps} caveat and consider the Umbanhowar–Goldman depth-dependent drag before proposing new terms. If the compensated plot is not flat, pursue a splay-angle-dependent exponent (à la Williams 2025's $\tan \phi$ crossover). If fitted k tracks convex-hull area but C tracks bare area (or vice versa), that split is the headline result.

Consolidated checklist — mapping your 5-step plan to deliverables

Step	Deliverable(s)	Figures	Tables	Fitted quantities	Key convention/citation
1. Audit + velocity binning	Single pooled v_0 range in text; per-height mean v_0	(none new; feeds all x-axes)	Supp. table: per-geometry v_0 range if requested	per-height mean v_0 , ± 1 SD	Measured v_0 from imaging, not $v(2gh)$ (KD 2007); SD not SE (Williams 2025)
2. Kinematics	1×3 (or 3×3) KD-style panels per geometry; insets deferred to after Step 5	Fig. 1: $z(t)$, $v(t)$, $a(t)+g$ per geometry, drop heights color-coded	—	(qualitative: t_{stop} , a_{stop} visible)	$a+g$ net accel (KD 2007 Fig. 1); own sign convention, depth positive into bed; $\bullet/\square/\triangle$ colorblind-safe markers
3. Area verification	Justified area convention per term	(schematic of convex hull vs bare area)	Table: convex-hull A, bare A, gap/grain ratio per geometry	k normalized by convex-hull A; C reported under both A	Convex hull for k (Kang et al. 2018); bare area for C (RFT, Li et al. 2013); free flow (Pravin et al. 2021)
4. Scaling	2/3-law test on linear axes; own exponent vs 2/3; area-resolved geometry comparison	Fig. 2a: d vs $v_0 + 2/3$ curves (n [CI] quoted); Fig. 2b: area-normalized $d/A^{1/3}$ vs v_0 (hull vs bare); Fig. 2c: collapse d/d_0 vs v_0/v^* ; (compensated plot optional/supp.)	Table: d_0 , β , n [CI] per geometry	d_0 , prefactor β (2/3 fixed); free n [CI] as cross-check	Linear axes, not log-log (Ambroso 2005; Clauset 2009); area link to Item 3
5. Dynamic force law	KD decomposition per geometry; add deferred insets here if desired	Fig. D1: $a+g$ vs v (inset v^2) fixed depths; Fig. D2: $F(z)/m$ collapse; Fig. D3: k , C , a_{stop} across geometries	Table: k , d_1 ($\rightarrow C$), a_{stop} slope per geometry, with separability flag	k , d_1 , C , $\Delta a = g + (k/m)d$	Single (k, d_1) per geometry across all speeds (KD 2013); separability check at φ (U&G 2010)

Caveats

Power-law over small range: with ~1–3 cm depths (<1 decade), any fitted exponent has large real uncertainty; report the free exponent with its full CI as a consistency check against the fixed 2/3 target ("2/3

in Cl''), not as a precision exponent measurement (Clauset, Shalizi & Newman 2009).

Area ambiguity: the depth (k) and inertial (C) terms may legitimately reference different effective areas; do not force one area on both. The free-flow-through-gaps conclusion is inferred by convergence of the parallel-intruder, clogging, and proppant-bridging literatures (Pravin et al. 2021; Zuriguel et al. 2005; Gruesbeck & Collins 1982) — no single paper tests rakes/combs at 50–100 grain-diameter gaps, so present it as a well-supported inference.

Packing fraction: clean additive KD separability is guaranteed only near ϕ_{cps} (Umbanhowar & Goldman 2010); verify at your $\phi = 0.624$ via the Fig. D1 depth-independence check.

Imaging vs accelerometer: optical differentiation cannot resolve fast force-chain transients; your $a(t)$ is a smoothed estimate (KD-style windowed fitting), adequate for k and d_i but not for peak-force fluctuation statistics.

Marker overlays: the $\otimes$ (circle-with- $\times$) symbol degrades in grayscale print and can be confused with a filled circle; prefer three distinct shapes.

Sign/normalization conventions: plot $a+g$ to match KD, but keep your own sign convention (depth positive into the bed); state it explicitly in captions with a one-line note mapping to KD's upward-positive z so readers can compare figures directly.